Construction of Equivalence

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1 Introduction (ND)

Fix G as a finite group. Let P_G be the *associativity G-operad* in $\mathbf{Set}_G$, defined by $P_G(n) := \mathbf{Set}(G, \Sigma_n)$, with structure maps defined by post composition

$$\gamma : \mathbf{Set}(G, \Sigma_k) \times \prod_{r=1}^{k} \mathbf{Set}(G, \Sigma_{j_r}) \cong \mathbf{Set}\left(G, \Sigma_k \times \prod_{r=1}^{k} \Sigma_{j_r}\right) \xrightarrow{\gamma \circ -} \mathbf{Set}(G, \Sigma_j),$$

, where $j = \sum_{r=1}^{k} j_r$, and $\gamma : \Sigma_k \times \prod_{r=1}^{k} \Sigma_{j_r} \to \Sigma_j$ is the structure map for associativity operad $\mathscr{M}$ in $\mathbf{Set}$. This is naturally a free $(G \times \Sigma)$-operad, as the right $(G \times \Sigma_n)$-action at each $P_G(n)$ is free.

However, it is mentioned in [1] that such G-operad is not finitely generated, as a rough explanation, for every $n \in \mathbb{N}$, there exists $m > n$ and a function $f : G \to \Sigma_m$, such that it cannot be generated using $\{P_G(k)\}_{k \leq n}$ with the operadic structure map and the $(G \times \Sigma)$-actions.

As a substitution, [1] defines a finitely-generated suboperad $Q_{G,\mathfrak{D}} \hookrightarrow P_G$ (for some parameter $\mathfrak{D}$ that's associated with choice of transitive G-sets), such that after chaotic generalization the two are equivalent in $\mathbf{Cat}_G$, and having equivalent categories of operadic algebras.

We'll mimic the statement, define an intermediate operad $Q_{G,\mathfrak{D}} \hookrightarrow Q_G \hookrightarrow P_G$, promote it to $\mathscr{Q}_G$ in $\mathbf{Cat}_G$, and *attempt to* modify the proof in [2] to show the equivalence of 2-categories between $\mathscr{Q}_G$-$\mathbf{PsAlg}$ and $\mathscr{F}_{G,\mathbb{R}}$-$\mathbf{PsAlg}$.

Future task (assume this works): Prove the equivalence of operads/algebras/pseudoalgebras with the original permuativity G-operad.

Current Goal: Add in extra axioms, so that there exist two 2-functors

$$\mathbb{R}_G : \mathscr{Q}_G\text{-}\mathbf{PsAlg} \leftrightarrows \mathscr{F}_{G,\mathbb{R},\_}\text{-}\mathbf{PsAlg} : \mathbb{Q}_G$$

inducing an isomorphism of (strict) 2-categories. Here, _ is some (yet to be determined) axiom.

2 Preliminaries (ND)

2.1 Symmetric Monoidal Categories

Definition 2.1. A category $\mathcal{V}$ is a *monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (\_ \otimes \_) \otimes \_ \xrightarrow{\sim} \_ \otimes (\_ \otimes \_)$ of functors $\mathcal{V} \times \mathcal{V} \times \mathcal{V} \xrightarrow{\sim} \mathcal{V}$. (So, $(X \otimes Y) \otimes Z \cong X \otimes (Y \otimes Z)$ in a canoncial way).
- The *left unit* $\lambda : \kappa \otimes \_ \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ ($\kappa \otimes X \cong X$ in a canonical way).
- The *right unit* $\rho : \_ \otimes \kappa \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ ($X \otimes \kappa \cong X$ in a canonical way).

They also satisfy two extra axioms:

- *Triangle Axiom:*

$$\begin{tikzcd}
(X \otimes \kappa) \otimes Y \arrow[rr, "a_{X,\kappa,Y}"] \arrow[dr, "\rho_X \otimes \mathrm{id}_Y"'] & & X \otimes (\kappa \otimes Y) \arrow[dl, "\mathrm{id}_X \otimes \lambda_Y"] \\
& X \otimes Y &
\end{tikzcd}$$

- *Pentagon Axiom:*

$$\begin{tikzcd}
((X \otimes Y) \otimes Z) \otimes W \arrow[rr, "a_{X \otimes Y,Z,W}"] \arrow[d, "a_{X,Y,Z} \otimes \mathrm{id}_W"'] & & (X \otimes Y) \otimes (Z \otimes W) \arrow[d, "a_{X,Y,Z \otimes W}"] \\
(X \otimes (Y \otimes Z)) \otimes W \arrow[dr, "a_{X,Y \otimes Z,W}"'] & & X \otimes (Y \otimes (Z \otimes W)) \\
& X \otimes ((Y \otimes Z) \otimes W) \arrow[ur, "\mathrm{id}_X \otimes a_{Y,Z,W}"'] &
\end{tikzcd}$$

Based on *Maclane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \to P_2$, where P_i are (possibly) different orders of performing product on $X_1, ..., X_n \in \mathcal{V}$. For reference, cf. *Category Theory for Working Mathematicians.*

This guarantees all monoidal category to be equivalent to a *strict monoidal category*,namely all a, λ, ρ above are "identity transformations", or $1 \otimes X = X, X \otimes 1 = X$, and $(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$, so WLOG one can assume the product $\otimes$ is associative.

Definition 2.2. A monoidal category $\mathcal{V}$ is *braided*, if there is a natural isomorphism $B : \_{}_1 \otimes \_{}_2 \xrightarrow{\sim} \_{}_2 \otimes \_{}_1$ (so $B_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X$ is an isomorphism), together with the *Hexagon axioms* satisfied:

$$\begin{tikzcd}
(X \otimes Y) \otimes Z \arrow[r, "a_{X,Y,Z}"] \arrow[d, "B_{X,Y} \otimes \mathrm{Id}_Z"'] & X \otimes (Y \otimes Z) \arrow[r, "B_{X,Y \otimes Z}"] & (Y \otimes Z) \otimes X \arrow[d, "a_{Y,Z,X}"] \\
(Y \otimes X) \otimes Z \arrow[r, "a_{Y,X,Z}"'] & Y \otimes (X \otimes Z) \arrow[r, "\mathrm{id}_Y \otimes B_{X,Z}"'] & Y \otimes (Z \otimes X)
\end{tikzcd}$$

$$\begin{tikzcd}
X \otimes (Y \otimes Z) \arrow[d, "\mathrm{id}_X \otimes B_{Y,Z}"'] & (X \otimes Y) \otimes Z \arrow[l, "a_{X,Y,Z}"'] \arrow[r, "B_{X \otimes Y,Z}"] & Z \otimes (X \otimes Y) \\
X \otimes (Z \otimes Y) & (X \otimes Z) \otimes Y \arrow[l, "a_{X,Z,Y}"] \arrow[r, "B_{X,Z} \otimes \mathrm{id}_Y"'] & (Z \otimes X) \otimes Y \arrow[u, "a_{Z,X,Y}"']
\end{tikzcd}$$

$\mathcal{V}$ is a *symmetric monoidal category*, if $B_{Y,X} = B_{X,Y}^{-1}$ for all $X, Y \in \mathcal{V}$.

The two hexagons are uniqueness of 3-cycle permutations on three consecutive objects in a product, hence together with B serving as transpositions, it guarantees unique isomorphism between all permutations of finite products, i.e., it imposes that there is a single way of doing such permutation (so there's no "nontrivial braiding", when $B_{Y,X} \circ B_{X,Y} : X \otimes Y \to Y \otimes X \to X \otimes Y$ is not identity). From my naive understanding of coherence theorem, WLOG we can again treat $X \otimes Y = Y \otimes X$.

2.2 Operads, Operadic Algebras

Definition 2.3. Given symmetric monoidal category $\mathcal{V}$, a Σ *operad* $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j\in\mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j-action (a group homomorphism $\Sigma_j^{\mathrm{op}} \to \mathrm{Aut}_{\mathcal{V}}(\mathcal{C}(j))$), a *unit map* $\eta : 1 \to \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes ... \otimes \mathcal{C}(j_k) \to \mathcal{C}(j_1 + ... + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{tikzcd} \mathcal{C}(j) \otimes \kappa^{\otimes j} \arrow[rr, "\sim"] \arrow[dr, "\mathrm{id}_{\mathcal{C}(j)} \otimes \eta^{\otimes j}"'] & & \mathcal{C}(j) \\ & \mathcal{C}(j) \otimes \mathcal{C}(1)^{\otimes j} \arrow[ur, "\gamma"'] & \end{tikzcd}, \qquad \begin{tikzcd} \kappa \otimes \mathcal{C}(j) \arrow[rr, "\sim"] \arrow[dr, "\eta \otimes \mathrm{id}_{\mathcal{C}(j)}"'] & & \mathcal{C}(j) \\ & \mathcal{C}(1) \otimes \mathcal{C}(j) \arrow[ur, "\gamma"'] & \end{tikzcd}$$

- *Associativity:* Fix natural numbers k, $j = \sum_{s=1}^k j_s$, define index $g_s := j_1 + ... + j_s$ for $1 \le s \le k$. For each $i_1, ..., i_{g_k} = i_j$, with sum $i = \sum_{r=1}^j i_r$, and $h_s := i_{g_{(s-1)}+1} + ... + i_{g_s}$ (so $i = \sum_{s=1}^k h_s$ also), the following diagram commutes:

$$\begin{tikzcd} \left[\mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(j_s)\right)\right] \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r)\right) \arrow[r, "\gamma\otimes\mathrm{id}"] \arrow[dd, "\text{shuffle}"'] & \mathcal{C}(j) \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r)\right) \arrow[d, "\gamma"] \\ & \mathcal{C}(i) \\ \mathcal{C}(k) \otimes \left[\bigotimes_{s=1}^k \left(\mathcal{C}(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} \mathcal{C}(i_{g_{(s-1)}+q})\right)\right)\right] \arrow[r, "\mathrm{id}_{\mathcal{C}(k)} \otimes (\bigotimes_s \gamma)"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(h_s)\right) \arrow[u, "\gamma"'] \end{tikzcd}$$

- *Equivariance:*
  - Given $\sigma \in \Sigma_k$, $j_1 + ... + j_k = \Sigma_j$, let $\sigma(j_1, ..., j_k) \in \Sigma_j$ denotes "permutation of k-blocks of letters", by $(j_1, ..., j_k) \mapsto (j_{\sigma^{-1}(1)}, ..., j_{\sigma^{-1}(k)})$.

    Example 2.4. Given $j = 5 = 2 + 3$, the permutation $(1\ 2) \in \Sigma_2$, $\sigma(2, 3)$ is the permutation $\langle 1, 2; 3, 4, 5\rangle \mapsto \langle 3, 4, 5; 1, 2\rangle$ (namely changing the blocks' order, without changing the order in each block).

Then, the following diagram commutes:

$$\begin{tikzcd}
\mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^{k} \mathcal{C}(j_r)\right) \arrow[r, "\sigma\otimes\, \mathrm{shuffle}_{\sigma^{-1}}"] \arrow[d, "\gamma"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^{k} \mathcal{C}(j_{\sigma(r)})\right) \arrow[d, "\gamma"] \\
\mathcal{C}(j) \arrow[r, "\sigma(j_{\sigma(1)},\ldots,j_{\sigma(k)})"'] & \mathcal{C}(j)
\end{tikzcd}$$

– Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus \ldots \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_rth block in j (for instance, let τ_1 acts on $1, \ldots, j_1$; let τ_2 acts on $j_1+1, \ldots, j_1+j_2$, etc.). Then, the following diagram commutes:

$$\begin{tikzcd}
\mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^{k} \mathcal{C}(j_r)\right) \arrow[r, "\mathrm{id}_{\mathcal{C}(k)}\otimes(\bigotimes_r \tau_r)"] \arrow[d, "\gamma"'] & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^{k} \mathcal{C}(j_r)\right) \arrow[d, "\gamma"] \\
\mathcal{C}(j) \arrow[r, "\tau_1\oplus\ldots\oplus\tau_k"'] & \mathcal{C}(j)
\end{tikzcd}$$

If $\mathcal{C}(0) = \kappa$ (with $\gamma : \mathcal{C}(0) \to \mathcal{C}$ being $\mathrm{id}_{\mathcal{C}(0)}$), the operad is *reduced*.

Definition 2.5. Let $\mathcal{C}, \mathcal{C}'$ be two operads on symmetric monoidal category $\mathcal{V}$, a *morphism of operads* $\psi : \mathcal{C} \to \mathcal{C}'$, is a sequence of Σ_j-equivariant morphisms $\psi_j : \mathcal{C}(j) \to \mathcal{C}'(j)$ in $\mathcal{V}$ (i.e., $\sigma^{-1} \circ \psi_j \circ \sigma = \psi_j$, for all $\sigma \in \Sigma_j$), such that it preserves the operad's structure morphism:

$$\begin{tikzcd}
\mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \ldots \otimes \mathcal{C}(j_k) \arrow[r, "\gamma"] \arrow[d, "\psi_k\otimes\psi_{j_1}\otimes\ldots\otimes\psi_{j_k}"'] & \mathcal{C}(j_1 + \ldots + j_k) \arrow[d, "\psi_{j_1+\ldots+j_k}"] \\
\mathcal{C}'(k) \otimes \mathcal{C}'(j_1) \otimes .. \otimes \mathcal{C}'(j_k) \arrow[r, "\gamma'"'] & \mathcal{C}'(j_1 + \ldots + j_k)
\end{tikzcd}$$

A primary example is the *associativity operad* in **Set** or $\mathscr{U}$ (the category of spaces):

Example 2.6. Fill in Details

Definition 2.7. Given an operad $\mathcal{C}$ over a symmetric monoidal category $\mathcal{V}$, a $\mathcal{C}$*-algebra* is an object $A \in \mathcal{V}$, with collections of morphisms $\theta : \mathcal{C}(j) \otimes A^j \to A$, that satisfies the following axioms:

- *Associativity:* Given $j = \sum_{i=1}^{k} j_i$, the following diagram commutes:

$$\begin{tikzcd}
\left[C(k) \otimes \left(\bigotimes_{i=1}^{k} C(j_i)\right)\right] \otimes A^{\otimes j} \arrow[r, "\gamma\otimes\mathrm{id}_{A^{\otimes j}}"] \arrow[dd, "\mathrm{shuffle}"'] & C(j) \otimes A^{\otimes j} \arrow[dr, "\theta"] & \\
 & & A \\
C(k) \otimes \left[\bigotimes_{i=1}^{k} \left(C(j_i) \otimes A^{\otimes j_i}\right)\right] \arrow[r, "\mathrm{id}_{C(k)}\otimes\theta^k"'] & C(k) \otimes A^{\otimes k} \arrow[ur, "\theta"'] &
\end{tikzcd}$$

(Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^{k} A^{\otimes j_i}$).

- *Unit:* The following diagram commutes:

$$\begin{tikzcd} \kappa \otimes A \arrow[rr, "\sim"] \arrow[dr, "\eta\otimes \mathrm{id}_A"'] & & A \\ & C(1) \otimes A \arrow[ur, "\theta"'] & \end{tikzcd}$$

- *Eauivariance:* Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{tikzcd} C(j) \otimes A^{\otimes j} \arrow[rr, "\sigma\otimes\ \mathrm{shuffle}_\sigma"] \arrow[dr, "\theta"'] & & C(j) \otimes A^{\otimes j} \arrow[dl, "\theta"] \\ & A & \end{tikzcd}$$

As an example, the operads over $\mathscr{M}$ mentioned previously is automatically a monoid (which describes the associativity structure).

2.3 2-Categories (ND)

2.4 Equivariance Preliminaries (ND)

2.5 G-Categories(ND)

3 Essential Categorical Tools (ND)

Here we introduce some definitions important to this paper.

3.1 G-Operad $\mathscr{Q}_G$ in $\mathbf{Cat}_G$

Definition 3.1. Let Q_G be the suboperad of P_G, generated by all the group homomorphisms $\alpha : G \to \Sigma_n$ spanning through all $n \in \mathbb{N}$, the P_G structure maps γ_G, and the $(G \times \Sigma)$-action on the elements. Define the operad $\mathscr{Q}_G := \mathscr{E}Q_G$ as the chaotic generalization in $\mathbf{Cat}_G$.

This operad is designed as a smallest suboperad that contains all group homomorphisms from G to a symmetric group, such that it's still a free $(G \times \Sigma)$-operad.

Potential (fun) question: Is this suboperad finitely generated, or equals to any $\mathscr{Q}_{G,\mathfrak{D}}$ described in [1]?

Warning 3.2. If naively taken all group homomorphisms and their orbits under $(G \times \Sigma)$-action, it doesn't form a suboperad in general for nontrivial group G. A counterexample can be constructed even for $G = \mathbb{Z}/2\mathbb{Z}$.

3.2 The 2-Category $\mathscr{Q}_G$-PsAlg

We mostly follow [2] in defining $\mathscr{Q}_G$-pseudoalgebras.

Definition 3.3 ($\mathscr{Q}_G$-pseudoalgebra)**.** A $\mathscr{Q}_G$-pseudoalgebra is a G-category $\mathscr{A}$ and operad action functors

$$\theta_j : \mathscr{Q}_G(j) \times \mathscr{A}^j \to \mathscr{A}$$

for each $n \in \mathbb{N}$, together with an invertible natural transformation ϕ_1 in the diagram

$$\begin{tikzcd} \mathscr{A} \arrow[r, "\iota"] \arrow[dr, "\mathrm{Id}_{\mathscr{A}}"'] & \mathscr{Q}_G(1) \times \mathscr{A} \arrow[d, "\theta_1"] \arrow[dl, phantom, "\Leftarrow \phi_1"] \\ & C \end{tikzcd} \tag{3.4}$$

and a natural transformation $\phi = \phi(k; j_1, ..., j_k)$ for each diagram

$$\begin{tikzcd} \mathscr{Q}_G(k) \times \left(\prod_{r=1}^k \mathscr{Q}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[rr, "\mathrm{id}\times\prod_r \theta_{j_r}"] \arrow[dd, "\mathrm{shuffle}"'] & & \mathscr{Q}_G(k) \times \mathscr{A}^k \arrow[d, "\theta_k"] \arrow[ddll, Rightarrow, "\phi"] \\ & & \mathscr{A} \\ \mathscr{Q}_G(k) \times \prod_{r=1}^k \mathscr{Q}_G(j_r) \times \mathscr{A}^{j_+} \arrow[rr, "\gamma\times\mathrm{id}"'] & & \mathscr{Q}_G(j_+) \times \mathscr{A}^{j_+} \arrow[u, "\theta_{j_+}"'] \end{tikzcd} \tag{3.5}$$

such that the following strict equivariance hold for θ_j and ϕ:

(i) **(Equivariance of θ_j)** The following diagram commutes for $(g, \sigma) \in G \times \Sigma_j$:

$$\begin{tikzcd} \mathscr{Q}_G(j) \times \mathscr{A}^j \arrow[rr, "(g,\sigma)\times(g,\sigma)^{-1}"] \arrow[dr, "\theta_j"'] & & \mathscr{Q}_G(j) \times \mathscr{A}^j \arrow[dl, "\theta_j"] \\ & \mathscr{A} \end{tikzcd} \tag{3.6}$$

(ii) **(Equivariance of ϕ)** The following equalities hold when given $g \in G$, $\sigma \in \Sigma_k$, and $\tau_r \in \Sigma_{j_r}$:

$$\phi(k; j_1, ..., j_k) = \phi(k; j_{\sigma(1)}, ..., j_{\sigma_k}) \circ ((g, \sigma) \times (g, \sigma)^{-1})$$

$$\phi(k; j_1, ..., j_k) = \phi(k; j_1, ..., j_k) \circ (\mathrm{id} \times \prod_r (\tau_r \times \tau_r^{-1}))$$

They also need to satisfy the following coherence properties:

(iii) **(Operadic Composition)** The following pasting diagrams are equal, given $j_+ = \sum j_r$, $i_r = \sum_s i_{r,s}$, and $i_+ = \sum_{r,s} i_{r,s}$ with $1 \le r \le k$ and $1 \le s \le j_r$:

$$\begin{tikzcd}
\mathscr{Q}_G(k) \times \prod_r \left(\mathscr{Q}_G(j_r) \times \prod_s \left(\mathscr{Q}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right)\right) \arrow[r, "\mathrm{id}\times\prod(\mathrm{id}\times\theta_{i_{r,s}})"] \arrow[dd, "\gamma\times\mathrm{id}"'] & Q_G(k) \times \prod_r \left(\mathscr{Q}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[dr, "\mathrm{id}\times\prod_{\theta_{j_r}}"] \arrow[dd, "\gamma"'] & \\
 & & \mathscr{Q}_G(k) \times \mathscr{A}^k \arrow[dl, Rightarrow, "\phi"] \arrow[dd, "\theta_k"] \\
\mathscr{Q}_G(j_+) \times \prod_{r,s} \left(\mathscr{Q}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right) \arrow[r, "\mathrm{id}\times\prod\theta_{i_{r,s}}"] \arrow[dr, "\gamma\times\mathrm{id}"'] & \mathscr{Q}_G(j_+) \times \mathscr{A}^{j_+} \arrow[d, Rightarrow, "\phi"] \arrow[dr, "\theta_{j_+}"] & \\
 & \mathscr{Q}_G(i_+) \times \mathscr{A}^{i_+} \arrow[r, "\theta_{j_+}"'] & \mathscr{A}
\end{tikzcd}$$

equals

$$\begin{tikzcd}
\mathscr{Q}_G(k) \times \prod_r \left(\mathscr{Q}_G(j_r) \times \prod_s \left(\mathscr{Q}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right)\right) \arrow[r, "\mathrm{id}\times\prod(\mathrm{id}\times\prod\theta_{i_{r,s}})"] \arrow[dr, "\mathrm{id}\times\prod(\gamma\times\mathrm{id})"'] \arrow[dd, "\gamma\times\mathrm{id}"'] & \mathscr{Q}_G(k) \times \prod_r \left(\mathscr{Q}_g(j_r) \times \mathscr{A}^{j_r}\right) \arrow[d, Rightarrow, "\mathrm{id}\times\prod\phi"] \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] & \\
 & \mathscr{Q}_G(k) \times \prod_r \left(\mathscr{Q}_G(i_r) \times \mathscr{A}^{i_r}\right) \arrow[r, "\mathrm{id}\times\prod\gamma"'] \arrow[dd, "\gamma\times\mathrm{id}"'] & \mathscr{Q}_G(k) \times \mathscr{A}^k \arrow[ddl, Rightarrow, "\phi"] \arrow[dd, "\theta_k"] \\
\mathscr{Q}_G(j_+) \times \prod_{r,s} \left(\mathscr{Q}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right) \arrow[dr, "\gamma\times\mathrm{id}"'] & & \\
 & \mathscr{Q}_G(i_+) \times \mathscr{A}^{i_+} \arrow[r, "\theta_{i_+}"'] & \mathscr{A}
\end{tikzcd}$$

(commuting squares are operadic associativity)

(iv) **(Operadic Identity)** The following two pairs of two pasting diagrams are equal:

$$\begin{tikzcd}
\mathscr{Q}_G(k) \times \mathscr{A}^k \arrow[r, "\mathrm{id}\times\iota^k"] & \mathscr{Q}_G(k) \times (\mathscr{Q}_G(1) \times \mathscr{A})^k \arrow[r, "\mathrm{id}\times\theta_1^k"] \arrow[d, "\mathrm{shuffle}"'] & \mathscr{Q}_G(k) \times \mathscr{A}^k \arrow[ddl, Rightarrow, "\phi"] \arrow[dd, "\theta_k"] \\
 & \mathscr{Q}_G(k) \times \mathscr{Q}_G(1)^k \times \mathscr{A}^k \arrow[d, "\gamma\times\mathrm{id}"'] & \\
 & \mathscr{Q}_G(k) \times \mathscr{A}^k \arrow[r, "\theta_k"'] & \mathscr{A}
\end{tikzcd}$$

equals

$$
\begin{tikzcd}
 & & \mathscr{Q}_G(k)\times(\mathscr{Q}_G(1)\times\mathscr{A})^k \arrow[d, phantom, "\Downarrow\, \mathrm{id}\times\prod\phi"] \arrow[dr, "\mathrm{id}\times\theta_1^k"] & & \\
\mathscr{Q}_G(k)\times\mathscr{A}^k \arrow[urr, "\mathrm{id}\times\iota^k"] \arrow[rrr, "\mathrm{id}"'] \arrow[d, "\mathrm{id}\times\iota^k"'] \arrow[dr, "\iota^k"] & & & \mathscr{Q}_G(k)\times\mathscr{A}^k \arrow[r, "\theta_k"] & \mathscr{A} \\
\mathscr{Q}_G(k)\times(\mathscr{Q}_G(1)\times\mathscr{A})^k \arrow[r, "\text{shuffle}"'] & \mathscr{Q}_G(k)\times\mathscr{Q}_G(1)^k\times\mathscr{A}^k \arrow[urr, "\gamma\times\mathrm{id}"'] & & &
\end{tikzcd}
$$

(uses operadic associativity)

Also:

$$
\begin{tikzcd}
\mathscr{Q}_G(k)\times\mathscr{A}^k \arrow[r, "\iota"] \arrow[dr, "\mathrm{id}"'] & \mathscr{Q}_G(1)\times\mathscr{Q}_G(k)\times\mathscr{A}^k \arrow[r, "\mathrm{id}\times\theta_k"] \arrow[d, "\gamma\times\mathrm{id}"'] & \mathscr{Q}_G(1)\times\mathscr{A} \arrow[d, "\theta_1"] \arrow[dl, Rightarrow, "\phi"] \\
 & \mathscr{Q}_G(k)\times\mathscr{A}^k \arrow[r, "\theta_k"'] & \mathscr{A}
\end{tikzcd}
$$

equals

$$
\begin{tikzcd}
\mathscr{Q}_G(k)\times\mathscr{A}^k \arrow[r, "\iota"] \arrow[d, "\theta_k"'] & \mathscr{Q}_G(1)\times\mathscr{Q}_G(k)\times\mathscr{A}^k \arrow[d, "\mathrm{id}\times\theta_k"] \\
\mathscr{A} \arrow[r, "\iota"] \arrow[dr, "\mathrm{id}"'] \arrow[r, phantom, shift right=3, "\Leftarrow\phi_1"'] & \mathscr{Q}_G(1)\times\mathscr{A} \arrow[d, "\theta_1"] \\
 & \mathscr{A}
\end{tikzcd}
$$

(Uses operadic unitality)

Definition 3.7 ($\mathscr{Q}_G$-pseudomorphism)**.** A $\mathscr{Q}_G$-pseudomorphism $(\mathbb{F},\zeta):(\mathscr{A},\theta,\phi)\to(\mathscr{B},\vartheta,\varphi)$ consists of *G-functor* $\mathbb{F}:\mathscr{A}\to\mathscr{B}$, such that for each j in the follogin diagram commutes up to an invertibal natural transformation ζ_j:

$$
\begin{tikzcd}
\mathscr{Q}_G(j)\times\mathscr{A}^j \arrow[r, "\mathrm{id}\times\mathbb{F}^j"] \arrow[d, "\theta_j"'] & \mathscr{Q}_G(j)\times\mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, Rightarrow, "\zeta_j"] \\
\mathscr{A} \arrow[r, "\mathbb{F}"'] & \mathscr{B}
\end{tikzcd}
$$

Moreover, they satisfy the following properties:

(i) **(Equivariance of ζ)** For each j,$\zeta_j=\zeta_j\circ((g,\sigma)\times(g,\sigma)^{-1})$ for all $(g,\sigma)\in G\times\Sigma_j$.

(ii) **(Preserves Identity)** The following two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{A} \arrow[r, "\mathbb{F}"] \arrow[d, "\iota"'] & \mathscr{B} \arrow[d, "\iota"'] \arrow[ddr, bend left, "\mathrm{id}"] & \\
\mathscr{Q}_G(1)\times\mathscr{A} \arrow[r, "\mathrm{id}\times\mathbb{F}"] \arrow[dr, "\theta_1"'] & \mathscr{Q}_G(1)\times\mathscr{B} \arrow[dr, "\vartheta_1"] \arrow[r, phantom, "\Rightarrow\varphi_1"] & {} \\
 & \mathscr{A} \arrow[r, "\mathbb{F}"'] \arrow[u, Rightarrow, "\zeta_1^{-1}"'] & \mathscr{B}
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\mathscr{A} \arrow[r, "\mathbb{F}"] \arrow[d, "\iota"'] \arrow[dr, "\mathrm{id}"] \arrow[d, phantom, shift left=4, "\Rightarrow\phi_1"] & \mathscr{B} \arrow[dr, "\mathrm{id}"] & \\
\mathscr{Q}_G(1)\times\mathscr{A} \arrow[r, "\theta_1"'] & \mathscr{A} \arrow[r, "\mathbb{F}"'] & \mathscr{B}
\end{tikzcd}
$$

(iii) **(Preserves Composition)** The following two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{Q}_G(k) \times \prod_r \left(\mathscr{Q}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[rrr, "\mathrm{id}\times\prod(\mathrm{id}\times\mathbb{F}^{j_r})"] \arrow[ddr, "\gamma"'] \arrow[drr, "\mathrm{id}\times\prod\theta_{j_r}"] & & & \mathscr{Q}_G(k) \times \prod_r \left(\mathscr{Q}_G(j_r) \times \mathscr{B}^{j_r}\right) \arrow[dr, "\mathrm{id}\times\prod\vartheta_{j_r}"] \arrow[dl, Rightarrow, "\mathrm{id}\times\prod\zeta_{j_r}"] & \\
 & & \mathscr{Q}_G(k) \times \mathscr{A}^k \arrow[rr, "\mathrm{id}\times\mathbb{F}^k"'] \arrow[dd, "\theta_k"'] \arrow[dl, Rightarrow, "\phi"] & & \mathscr{Q}_G(k) \times \mathscr{B}^k \arrow[dd, "\vartheta_k"] \arrow[ddll, Rightarrow, "\zeta_k"] \\
 & \mathscr{Q}_G(j) \times \mathscr{A}^j \arrow[dr, "\theta_j"'] & & & \\
 & & \mathscr{A} \arrow[rr, "\mathbb{F}"'] & & \mathscr{B}
\end{tikzcd}
$$

equals

$$
\begin{tikzcd}
\mathscr{Q}_G(k) \times \prod_r \left(\mathscr{Q}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[rr, "\mathrm{id}\times\prod(\mathrm{id}\times\mathbb{F}^{j_r})"] \arrow[dd, "\gamma"'] & & \mathscr{Q}_G(k) \times \prod \left(\mathscr{Q}_G(j_r) \times \mathscr{B}^{j_r}\right) \arrow[dd, "\gamma"'] \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] & \\
 & & & \mathscr{Q}_G(k) \times \mathscr{B}^k \arrow[dd, "\vartheta_k"] \arrow[dl, Rightarrow, "\varphi"] \\
\mathscr{Q}_G(j) \times \mathscr{A}^j \arrow[rr, "\mathrm{id}\times\mathbb{F}^j"] \arrow[dr, "\theta_j"'] & & \mathscr{Q}_G(j) \times \mathscr{B}^j \arrow[dr, "\vartheta_j"] \arrow[dl, Rightarrow, "\zeta_j"] & \\
 & \mathscr{A} \arrow[rr, "\mathbb{F}"'] & & \mathscr{B}
\end{tikzcd}
$$

Definition 3.8 ($\mathscr{Q}_G$-2-cells)**.** Let $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : (\mathscr{A}, \theta, \phi) \to (\mathscr{B}, \vartheta, \varphi)$ be two $\mathscr{Q}_G$-pseudomorphisms, a $\mathscr{Q}_G$-2-cell $\lambda : (\mathbb{F}, \zeta) \Rightarrow (\mathbb{G}, \xi)$ is a natural transformation $\lambda : \mathbb{F} \to \mathbb{G}$, such that the following two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{Q}_G(j) \times \mathscr{A}^j \arrow[r, bend left=50, "\mathrm{id}\times\mathbb{F}^j"] \arrow[r, bend right=50, "\mathrm{id}\times\mathbb{G}^j"'] \arrow[r, phantom, "\overset{\mathrm{id}\times\lambda^j}{\Longrightarrow}"] \arrow[d, "\theta_j"'] & \mathscr{Q}_G(j) \times \mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, Rightarrow, "\xi_j"] \\
\mathscr{A} \arrow[r, "\mathbb{G}"'] & \mathscr{B}
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\mathscr{Q}_G(j) \times \mathscr{A}^j \arrow[r, "\mathrm{id}\times\mathbb{F}^j"] \arrow[d, "\theta_j"'] & \mathscr{Q}_G(j) \times \mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, Rightarrow, "\zeta_j"'] \\
\mathscr{A} \arrow[r, bend left=40, "\mathbb{F}"] \arrow[r, bend right=40, "\mathbb{G}"'] \arrow[r, phantom, "\overset{\lambda}{\Longrightarrow}"] & \mathscr{B}
\end{tikzcd}
$$

3.3 Category $\mathscr{F}_G$ (ND)

3.4 2-Category $\mathscr{F}_G$-PsAlg, Version 1

In the category $\mathscr{F}_G$ (all based finite G-sets), the homsets $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ can also be viewed as a discrete category. We'll use this identification to define some notions.

Definition 3.9 ($\mathscr{F}_G$-pseudoalgebra)**.** An $\mathscr{F}_G$-pseudoalgebra, is a pseudofunctor $\tilde{\mathscr{A}} : \mathscr{F}_G \to \mathbf{Cat}_G$, such that restricting to the full subcategory $\Pi_G \subset \mathscr{F}_G$, $\tilde{\mathscr{A}}$ is a strict functor. Moreover, we require it to be reduced, namely $\tilde{\mathscr{A}}(0) = *$ (the terminal G-category), and each hom-functor

$$\tilde{\mathscr{A}}_{\mathbf{n}^\alpha,\mathbf{m}^\beta} : \mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) \to \mathbf{Cat}_G(\tilde{\mathscr{A}}(\mathbf{n}^\alpha), \tilde{\mathscr{A}}(\mathbf{m}^\beta))$$

is a G-functor.

Question: In operad case, we have natural isomorphism between any two G-actions, do we need it here?

Let $\delta_i : \mathbf{n}^\alpha \to \mathbf{1}$ be the ith projection (namely $i \mapsto 1$, and any $i \neq j \mapsto 0$), since it belongs to Π_G, it generates a functor $\tilde{\mathscr{A}}\delta_i : \tilde{\mathscr{A}}(\mathbf{n}^\alpha) \to \tilde{\mathscr{A}}(\mathbf{1})$. Hence, by universality of product, all $\tilde{\mathscr{A}}\delta_i$ $(1 \leq i \leq n)$ induces a functor $\delta : \mathring{\tilde{A}}(\mathbf{n}^\alpha) \to \tilde{A}(\mathbf{1})^n$.

Definition 3.10 ($\mathscr{F}_{G,\mathrm{vs}}$ and $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra)**.** An $\mathscr{F}_G$-pseudoalgebra $\tilde{\mathscr{A}}$ is *very special* (or an $\mathscr{F}_{G,\mathrm{vs}}$-pseudoalgebra), if all $\delta : \tilde{\mathscr{A}}(\mathbf{n}^\alpha) \to \tilde{\mathscr{A}}(\mathbf{1})^{\mathbf{n}^\alpha}$ is an isomorphism for all $\mathbf{n}^\alpha \in \mathscr{F}_G$, and $\tilde{\mathscr{A}}$ is *strictly special* (or an $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra), if all δ is in fact identity.

Definition 3.11 ($\mathscr{F}_G$-1-cell)**.** Given $\tilde{A}, \tilde{B}$ two $\mathscr{F}_G$-pseudoalgebras, a morphism (or $\mathscr{F}_G$-1-cell) between them is a pseudo-transformation $(\mathbb{F}, \zeta) : \tilde{\mathscr{A}} \to \tilde{\mathscr{B}}$, such that restricting to Π_G, it becomes an actual natural transformation.

Fill in details for the commutativity diagrams

Definition 3.12 ($\mathscr{F}_G$-2-cell)**.** Given $\tilde{A}, \tilde{B}$ two $\mathscr{F}_G$-pseudoalgebras, and two $\mathscr{F}_G$-1-cells $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : \tilde{A} \to \tilde{\mathscr{B}}$, an $\mathscr{F}_G$-2-cell is a transformation $\lambda : \mathbb{F} \Rightarrow \mathbb{G}$, such that for any morphism $f : \mathbf{n}^\alpha \to \mathbf{m}^\beta$ in $\mathscr{F}_G$, the following two pasting diagrams are equal:

$$\begin{tikzcd}
\tilde{\mathscr{A}}(\mathbf{n}^\alpha) \arrow[r, bend left=40, "\mathbb{F}_{\mathbf{n}^\alpha}"] \arrow[r, Rightarrow, "\lambda"] \arrow[r, bend right=40, "\mathbb{G}_{\mathbf{n}^\alpha}"'] \arrow[d, "\tilde{\mathscr{A}}f"'] & \tilde{\mathscr{B}}(\mathbf{n}^\alpha) \arrow[d, "\tilde{\mathscr{B}}f"] \arrow[dl, Rightarrow, "\xi f"] \\
\tilde{A}(\mathbf{m}^\beta) \arrow[r, "\mathbb{G}_{\mathbf{m}^\beta}"'] & \tilde{B}(\mathbf{m}^\beta)
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\tilde{\mathscr{A}}(\mathbf{n}^\alpha) \arrow[r, "\mathbb{F}_{\mathbf{n}^\alpha}"] \arrow[d, "\tilde{\mathscr{A}}f"'] & \tilde{\mathscr{B}}(\mathbf{n}^\alpha) \arrow[d, "\tilde{\mathscr{B}}f"] \arrow[dl, Rightarrow, "\zeta f"'] \\
\tilde{A}(\mathbf{m}^\beta) \arrow[r, bend left=40, "\mathbb{F}_{\mathbf{m}^\beta}"] \arrow[r, Rightarrow, "\lambda"'] \arrow[r, bend right=40, "\mathbb{G}_{\mathbf{m}^\beta}"'] & \tilde{B}(\mathbf{m}^\beta)
\end{tikzcd}$$

4 Construction of $\mathbb{R}_G$ (ND)

5 Counterexample: $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra that's not $\mathscr{Q}_G$-pseudoalgebra

Here we fix the group $G := \mathbb{Z}/2\mathbb{Z}$. Which, we have $\mathscr{Q}_G(1) = \mathscr{P}_G(1) = *$, in particular $G \times \Sigma_1 \cong G$ has trivial action on these. This leads to the following properties of $\mathscr{Q}_G$ or $\mathscr{P}_G$-pseudoalgebra:

Proposition 5.1. *For any $\mathscr{Q}_G$-pseudoalgebra $\mathscr{A}$, any G-action $g : \mathscr{A} \xrightarrow{\sim} \mathscr{A}$ is naturally isomorphic to the identity functor $\mathrm{Id}_{\mathscr{A}}$.*

Proof. Pasting the unitality and equivariance diagram together, we get the following naturality diagram:

$$\begin{tikzcd}
\mathscr{A} \arrow[rrr, "g^{-1}"] \arrow[dr, "\iota"] \arrow[ddr, bend right, "\mathrm{Id}_{\mathscr{A}}"'] & & & \mathscr{A} \arrow[dl, "\iota"'] \arrow[ddl, bend left, "\mathrm{Id}_{\mathscr{A}}"] \\
& \mathscr{Q}_G(1)\times\mathscr{A} \arrow[r, "g\times g^{-1}"] \arrow[d, "\theta_1"] \arrow[l, phantom, "\Leftarrow \phi_1"] & \mathscr{Q}_G(1)\times\mathscr{A} \arrow[d, "\theta_1"] & \arrow[l, phantom, "\Leftarrow \phi_1^{-1}"] \\
& \mathscr{A} \arrow[r, equal] & \mathscr{A} &
\end{tikzcd} \tag{5.2}$$

Here, the top trapezoid commutes because G acts trivially on $\mathscr{Q}_G(1)$. This identifies a natural isomorphism as follow:

$$g^{-1} = \mathrm{Id}_{\mathscr{A}} \circ g^{-1} \xRightarrow{\phi_1^{-1} * \mathrm{id}_{g^{-1}}} (\theta_1 \circ \iota) \circ g^{-1} = \theta_1 \circ (g \times g^{-1}) \circ \iota = \theta_1 \circ \iota \xRightarrow{\phi_1} \mathrm{Id}_{\mathscr{A}} \tag{5.3}$$

$\square$

To construct a counterexample for our fixed G, it suffices to find an $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra that doesn't have the above property. Here, we largely use the fact that for two naturally isomorphic functors, if the target category is discrete, the two functors must be equal.

Construction 5.4. Let $\mathscr{A} := \{0, 1\}$ denotes the discrete category with 2 objects, which a G-action on $\mathscr{A}$ is specified by its action on the objects due to its discreteness. Define a G-action on $\mathscr{A}$ given by

$$H_0 : \mathscr{A} \xrightarrow{\sim} \mathscr{A}, \quad H_0(0) = 0,\ H_0(1) = 1 \tag{5.5}$$
$$H_1 : \mathscr{A} \xrightarrow{\sim} \mathscr{A}, \quad H_1(0) = 1,\ H_1(1) = 0, \tag{5.6}$$

Which, $H_0^2 = H_0$, $H_0 \circ H_1 = H_1 \circ H_0$, and $H_1^2 = H_0$, so H_0, H_1 satisfies the relations of $G = \mathbb{Z}/2\mathbb{Z}$, hence inducing a G-action on $\mathscr{A}$.

Define $\tilde{\mathscr{A}} : \mathscr{F}_G \to \mathbf{Cat}_G$ given by $\tilde{\mathscr{A}}(\mathbf{n}^\alpha) := \mathscr{A}^{\mathbf{n}^\alpha}$ (with underlying category $\mathscr{A}^n$, a discrete category with 2^n objects), and for any morphism $f : \mathbf{n}^\alpha \to \mathbf{m}^\beta$, define $\tilde{\mathscr{A}} f : \mathscr{A}^{\mathbf{n}^\alpha} \to \mathscr{A}^{\mathbf{m}^\beta}$ by

$$\tilde{\mathscr{A}} f\left((a_i)_{i=1,\ldots,n}\right) := \left(\sum_{i \in f^{-1}(j)} a_i \bmod 2\right)_{j=1,\ldots,m}, \tag{5.7}$$

for special case where $f^{-1}(j) = \emptyset$, define the jth entry as 0.

To check it preserves composition up to coherent natural isomorphism, consider $\mathbf{n}^\alpha \xrightarrow{f} \mathbf{m}^\beta \xrightarrow{g} \mathbf{k}^\gamma$ in $\mathscr{F}_G$, then $\tilde{\mathscr{A}}(g \circ f) : \tilde{\mathscr{A}}^{\mathbf{n}^\alpha} \to \tilde{\mathscr{A}}^{\mathbf{m}^\beta}$ is given as follow:

$$\tilde{\mathscr{A}}(g \circ f)((a_i)_{i=1,\dots,n}) = \left(\sum_{i \in (g\circ f)^{-1}(\ell)} a_i \bmod 2 \right)_{\ell=1,\dots,k} \tag{5.8}$$

$$= \left(\sum_{i \in f^{-1}(g^{-1}(\ell))} a_i \bmod 2 \right)_{\ell=1,\dots,k} \tag{5.9}$$

$$= \left(\sum_{j \in g^{-1}(\ell)} \left(\sum_{i \in f^{-1}(j)} a_i \bmod 2 \right) \bmod 2 \right)_{\ell=1,\dots,k} \tag{5.10}$$

$$= \tilde{\mathscr{A}} g \left(\left(\sum_{i \in f^{-1}(j)} a_i \bmod 2 \right)_{j=1,\dots,m} \right) \tag{5.11}$$

$$= \tilde{\mathscr{A}} g \left(\tilde{\mathscr{A}} f((a_i)_{i=1,\dots,k}) \right) \tag{5.12}$$

(Here, the well-defineness of the function heavily relies on the assumption that if $f^{-1}(j) = \emptyset$, the corresponding sum is 0). In particular, $\tilde{\mathscr{A}}(g \circ f) = \tilde{\mathscr{A}} g \circ \tilde{\mathscr{A}} f$, so $\tilde{\mathscr{A}}$ is a functor (or an $\mathscr{F}_G$-algebra), and since each $\tilde{\mathscr{A}}(\mathbf{n}^\alpha) := \mathscr{A}^{\mathbf{n}^\alpha}$, it's in fact an $\mathscr{F}_{G,\mathbb{R}}$-algebra.

Yet, since $\mathscr{A}$ is discrete, with H_0, H_1 being distinct functors on $\mathscr{A}$, we see that not all the G-actions on $\mathscr{A}$ are naturally isomorphic to $\mathrm{Id}_{\mathscr{A}}$, violating the property described in previous proposition. Hence, $\mathscr{A}$ cannot be a $\mathscr{Q}_G$-pseudoalgebra, or even a $\mathscr{P}_G$-pseudoalgebra due to the same reasoning.

This shows that the functor $\mathbb{R}_G : \mathscr{P}_G$-**PsAlg** $\to \mathscr{F}_G$-**PsAlg** is not essentially surjective, hence cannot be an equivalence of categories. Same statement holds true for $\mathscr{Q}_G$-**PsAlg**.

5.1 Extra Axioms Required on $\mathscr{F}_G$-PsAlg

Possible things to assume: the 1st-space $\mathscr{A} := \mathscr{B}(\mathbf{1})$ should have a *chaotic G-actions*, meaning any $g \in G$ should describe an isomorphism functor $g : \mathscr{A} \xrightarrow{\sim} \mathscr{A}$, such that $g, \mathrm{Id}_{\mathscr{A}}$ are naturally isomorphic.

Question: This is necessary, but is it sufficient?

6 Construction of $\mathbb{Q}_G$ under Extra Axiom (ND)

7 (If everything works out) $\mathbb{R}_G$ and $\mathbb{Q}_G$ are Isomorphisms (ND)

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